

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Tech (ME-AMT) Examination

December 2017

ME763 Manufacturing Planning and Control

Date: 26.12.2017, Tuesday

Time: 10.00 a.m. To 01.00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1** How do you map matching the Manufacturing Planning & Control (MPC) System with the Needs of the Firm in changing world? **[07]**
- Q – 2.a** Materials Requirements Planning (MRP) and Just-in-Time (JIT) are currently two of the most popular manufacturing technologies It has been found that the degree of the MRP and JIT implementation and integration has a positive relationship with the manufacturer's performance. **“The hybrid MRP and JIT system, which create synergy and attains better performance.”** Give your comments on this statement. **[04]**
- Q – 2.b** Answer any two questions. **[10]**
- (i) What does the Toyota Production System accomplish?
 - (ii) Elaborate following elements of JIT that make this system run
 - 1 External Setup
 - 2 Flexible Changeover
 - 3 Heisenberg Principle
 - 4 Internal Setup
 5. Internal Customer
 6. Kanban
 - 7 Machine Cell
 - 8 Level Loading
 - 9 Poka-yoke
 - (iii) Elaborate following elements of Computer Aided Process Planning (CAPP)
 1. Process Planning
 2. Variant Process Planning
 - 3 Generative Process Planning
 4. Feature Recognition

Q - 3 Answer any one.**[14]**

- a. How would you use measure forecast accuracy? Can you suggest model in front of description in table 1

Description	Model
Uses last period's actual value as a forecast	-----
Uses an average of a specified number of the most recent observations, with each observation receiving the same emphasis (weight)	-----
Uses an average of a specified number of the most recent observations, with each observation receiving a different emphasis (weight)	-----
A weighted average procedure with weights declining exponentially as data become older	-----
Technique that uses the least squares method to fit a straight line to the data	-----
A mechanism for adjusting the forecast to accommodate any seasonal patterns inherent in the data	-----

Table 1

- b. Examines the current state of Production Planning and Control (PPC) and identified some technical and systems changes that have occurred over recent years and links these with the requirements of following terms
1. Administrative actions
 2. Behavioral actions
 3. Quantitative actions

SECTION – II

Q - 4 How would you clear to someone the following terms? [07]

- a.** Design For Manufacturing (DFM)
- b.** Design For Quality (DFQ)
- c.** Design For Reliability (DFS)
- d.** Design For Robustness (DFR)
- e.** Design For Optimization(DFO)
- f.** Design For Assembly(DFA)
- g.** Quality Function Deployment (QFD)

Q – 5.a Discuss the meaning, scope and motivation of Design of Experiments (DOE). [05]

OR

.Q – 5.a Explain types of experiments related with DOE. [05]

Q – 5.b Identify types of model in DOE. [04]

Q – 5.c Discuss types of designs in DOE. [05]

OR

Q – 5.c Can you write in your own words meaning of Analysis of Variance (ANOVA)? [05]

Q – 6.a What activities did you carried out on MINITAB software in your laboratory [05]
during last semester?

Q – 6.b Under what conditions would you recommend push and pull system? [05]

Q – 6.c Illustrate a way to carry out quality control in an organization? [04]

OR

Q – 6.a What can you interpret from control charts? [05]

Q – 6.b Justify your conclusion about “six sigma reduce the defects”. [05]

Q – 6.c Is it that hard for India to produce goods/services as low as it costs in China? [04]
We have people and cheap labor too. What is the factor that is keeping China
the No.1 manufacturer?